

Walking into the heat dome

Sixty-four GenCast walkers, propagated from 21 days before the 2021 Pacific Northwest heat dome and resampled five times on a FourCastNet 3 (FCN3) score. Forty-two of them reached the anomaly ERA5 recorded. None of forty free-running GenCast members did.

42/64 AI+RES walkers reaching the observed A_L	0/40 direct GenCast members reaching it	+7.72 K the ERA5 value over the PNW box	+10.29 K the most extreme walker of the run	7 h 45 m one 8×H100 node, end to end
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1 · THE RUN Every branch the pilot rolled, from the init to the peak

The observable A_L is the $\cos(\text{lat})$ -weighted mean, over a box covering Oregon and Washington, of the 7-day-mean 2 m temperature anomaly for the week ending at the event peak. The vertical axis of Fig. 1 is the same index evaluated frame by frame, that is, the value a walker holds at that moment rather than the value it will average to. It is plotted as a 24-hour running mean, because the raw 12-hourly index has a diurnal swing of several kelvin that on 64 lineages obscures the trend. The five dotted verticals mark the resampling barriers.

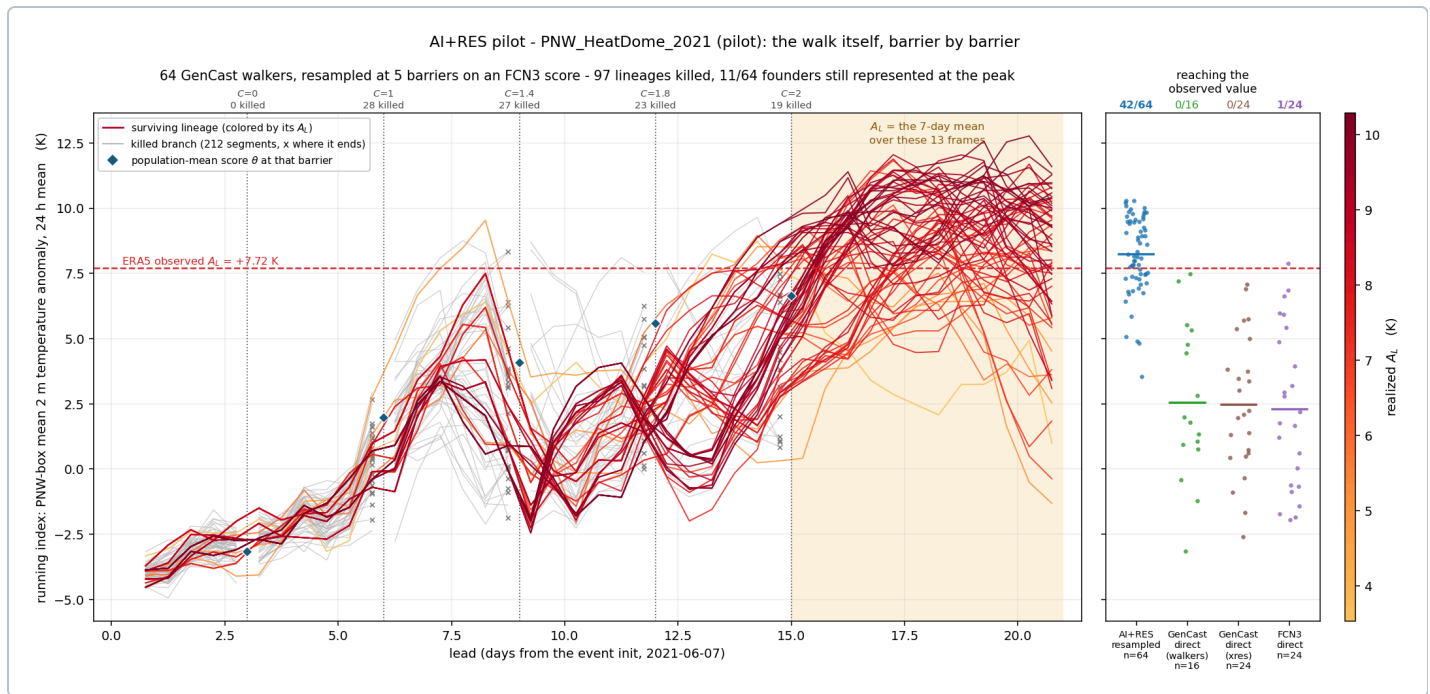


Figure 1. Every lineage of the pilot, init to peak. Colored lines are the 64 lineages that survive to the peak, tinted by the A_L each one ends with. Grey lines are branches killed at a barrier, each marked with a small x. The shaded band at the right is the 7-day window over which A_L is averaged. The right panel gives where every method's members landed on the same axis, against the ERA5 value.

HOW TO READ FIG. 1

- **A surviving lineage**, which is a chain of segments rather than one walker. Slot 3's state at 15 d and its state at 6 d can belong to different ancestries, so each trajectory is assembled by walking the parent table backwards from the peak.
- **A killed branch**. Its score at that barrier gave it no children in the resampling. The compute already spent on it is not wasted, since it is what told the scheme where *not* to look, but the branch ends there.
- **The population-mean score at each barrier**. θ is FCN3's 6-member forecast of the walker's final A_L , and it is the only quantity the resampling reads. It rises $+1.98 \rightarrow +4.07 \rightarrow +5.57 \rightarrow +6.62$ K across the four scored barriers.
- **The 7-day window**, lead 15 d to 21 d, thirteen 12-hourly frames ending at the peak. Everything to its left is the walk, and A_L is one average taken inside it. No resampling happens in the window, since the last barrier is at 15 d and the event week is a free 6-day run-out.
- **ERA5, +7.72 K**. The killed branches sit predominantly below it.

WHAT FIG. 1 DOES NOT TEST

Fig. 1 is not evidence that the score was right. Trajectories are drawn from the walkers' own diagnostics and the barriers are drawn where the schedule put them, so nothing in the figure tests whether θ predicted the outcome. That test is Fig. 8 of Section 7, and it was passed before this run was submitted, as Gate 3 of Table 7.

2 · SCHEDULE Five barriers, each one harder than the last

At barrier k every walker is scored, the scores are standardized across the live population, and each walker is cloned a number of times proportional to $\exp(C_k \cdot \text{standardized score})$. C_k is the selection pressure. At $C_k = 0$ the resampling does nothing, and the ramp was set so that the early barriers, where the score is weakest, remove the fewest walkers.

BARRIER 1		BARRIER 2		BARRIER 3		BARRIER 4		BARRIER 5	
3 d		6 d		9 d		12 d		15 d	
C_k	0	C_k	1.0	C_k	1.4	C_k	1.8	C_k	2.0
killed	0	killed	28	killed	27	killed	23	killed	19
ESS/N	1.00	ESS/N	0.28	ESS/N	0.26	ESS/N	0.46	ESS/N	0.60
max clones	1	max clones	11	max clones	11	max clones	8	max clones	3

Barrier 1 is inert by construction. At 3 d lead the score has no skill on this event, $\rho_s = -0.04$ as measured in Gate 3 and reported in Table 4, so resampling on it would select on noise. The population diversity it preserves is what the later barriers spend.

THE SCHEDULE WAS TUNED BEFORE THE RUN, AND HELD

Replaying the same C_k on Gate 3's measured skill curve predicted effective sample sizes of 1.00 / 0.43 / 0.36 / 0.38 / 0.58. The pilot delivered 1.00 / 0.28 / 0.26 / 0.46 / **0.60** against a target of $N/4 = 0.25$. Barriers 2 and 3 landed near the 5th percentile of the replay and barrier 5 landed at the predicted value. $C_k = (0, 1.0, 1.4, 1.8, 2.0)$ is confirmed at $N = 64$ on real scores.

The score, and the rule that decides who splits and who dies

1 • THE SCORE IS A FORECAST, NOT A MEASUREMENT

At barrier k , walker i 's global state is converted to a 72-channel FCN3 initial condition and rolled to the event peak M times. Its score, Eq. (1), is the mean of what those M members say the observable will be.

EQ. (1) • THE SCORE

$$\theta_k(i) = (1/M) \sum_{m=1..M} A_L(FCN3_m(\text{adapter}(X_k^i)))$$

X_k^i walker i 's global GenCast state at barrier k , 12 prognostic variables on 13 levels, two time frames.

$FCN3_m$ the m -th stochastic member, rolled at 6 h steps from the barrier to the peak. All M members share the same initial condition and differ only through FCN3's own noise generator.

$A_L(\cdot)$ the $\cos(\text{lat})$ -weighted PNW-box mean of the 7-day-mean anomaly for the week ending at the peak, the same function applied to the walkers themselves at the end of the run.

$M = 6$ so θ carries a sampling error of about 0.25 K on this event, which is enough to *rank* walkers and not enough to *equate* them.

θ is a prediction of where the walker is going, not a reading of where it is. The cheap alternative, scoring a walker by its current anomaly, is the Standard-RES of Lancelin et al., and Table 4 shows it has no skill at the leads that matter here.

2 • FROM SCORE TO SELECTION PRESSURE TO OFFSPRING COUNT

Scores in kelvin cannot drive a schedule that has to work across events with different units and spreads. The score is therefore standardized across the live population and scaled by the barrier's selection pressure, Eq. (2).

EQ. (2) • STANDARDIZED SCORE AND TILT

$$z_k(i) = (\theta_k(i) - \text{mean}_j \theta_k(j)) / \text{sd}_j \theta_k(j)$$

$$V_k(i) = C_k \cdot z_k(i)$$

A walker's weight is the **change** in v along the leg it just travelled, from its parent's value to its own. The population mean of those weights normalizes the leg, Eq. (3).

EQ. (3) • INCREMENTAL WEIGHT AND ITS MEAN

$$w_k(i) = \exp(V_k(i) - V_{k-1}(\text{parent of } i))$$

$$R_k = (1/N) \sum_i w_k(i)$$

EQ. (4) • EXPECTED NUMBER OF CHILDREN

$$n_k(i) = w_k(i) / R_k \quad \sum_i n_k(i) = N \text{ by construction}$$

Dividing by R_k holds the population at N . Cloning the good walkers is paid for by killing others, and that accounting is done in Eq. (4) rather than by capping or padding afterwards.

3 • FROM A FRACTIONAL CHILD COUNT TO AN INTEGER

$n_k(i)$ is a real number such as 4.17 or 0.21, and a walker cannot have 0.21 children.

Because n is often greater than 1, which is what cloning is, this is an allocation problem rather than a selection one.

- **Guaranteed children.** Every walker gets $\text{floor}(n_k(i))$ children outright. A walker with $n = 4.17$ is guaranteed 4.
- **Pivotal sampling allocates the leftover fractions** (Deville & Tillé, 1998). The fractions sum to a whole number, pivotal sampling hands out that many extra children, and walker i receives one with probability equal to its own fraction. It repeatedly takes two undecided walkers and moves probability mass between them until one is settled at 0 or 1, which preserves every marginal exactly and conserves the total at every move.

TWO ALTERNATIVES, AND WHY THEY FAIL

Independent coin flips on each fraction give the right per-walker probabilities and a fluctuating population size, 61 walkers one step and 67 the next, which makes every downstream budget wrong. A multinomial draw fixes the total and inflates the variance of the offspring counts, which is the noise the estimator has to fight. Pivotal sampling gives exact marginals and an exact total.

A walker is killed when it gets zero children, that is, when its floor is 0 and it loses the fractional draw. Nothing else kills a walker, and there is no threshold and no ranking cutoff. The offspring vector is then flattened into a parent list, so counts [0, 2, 0, 1] become parents [1, 1, 3], and the next leg rolls one segment per child from the named parent's state.

4 • THE ALLOCATION AT BARRIER 2 OF THIS RUN

Barrier 2 is the simplest case to read, because $c_1 = 0$ makes $v_1 \equiv 0$ and therefore $w_2(i) = \exp(z_2(i))$ exactly. The run used $c_2 = 1.0$ and measured $R_2 = 1.7702$.

Table 1 gives the allocation that follows.

z_2	$w_2 = e^z$	$N_2 = W/R$	GUARANTEED	P(ONE MORE)	P(KILLED)	OUTCOME
-2.0	0.135	0.076	0	0.076	0.924	almost certainly dies
-1.0	0.368	0.208	0	0.208	0.792	4-in-5 chance of dying
0.0	1.000	0.565	0	0.565	0.435	the <i>median</i> walker, and still a 0.435 chance of dying
+1.0	2.718	1.536	1	0.536	0	survives, coin-flip for a twin
+2.0	7.389	4.174	4	0.174	0	four clones guaranteed, sometimes five

Table 1. Offspring allocation at barrier 2. Every entry follows from $c_2 = 1$ and the measured $R_2 = 1.7702$. P(killed) is $1 - n$ whenever $n < 1$, because such a walker has no guaranteed child and its only chance is the fractional draw.

R_k is the mean of $\exp(z)$, and by Jensen's inequality that mean exceeds $\exp(\text{mean } z) = 1$. A walker sitting at the population mean score is therefore below the mean weight and at risk, with $P(\text{killed}) = 0.435$ in the third row of Table 1. That is why 28 of 64 walkers died at a barrier whose selection pressure was 1.0. The scheme does not preserve the middle of the population. R_k falls below 1 at later barriers, where v_{k-1} is no longer zero and w measures a change rather than a level.

5 • UNDOING THE TILT TO RECOVER A PROBABILITY

Because v depends only on state, the weights telescope along a lineage, and the product of w down any surviving branch is $\exp(v_k - v_0)$. After the last barrier the

population is distributed under the tilted law. Any expectation under the original law is recovered by dividing the tilt back out, Eq. (5).

EQ. (5) • THE ESTIMATOR

$$E_P[\phi] = Z \cdot (1/N) \sum_i \phi(X^i) \cdot \exp(-V_K(\text{lineage of } i)) \quad , \quad Z = \prod_k R_k$$

Setting $\phi = 1\{A_L \geq a\}$ in Eq. (5) gives the exceedance curve of Fig. 2. For this run $Z = 1.0000 \times 1.7702 \times 0.5468 \times 0.6787 \times 0.5995 = 0.39387$ and $\log Z = -0.9317$.

EQ. (6) • POPULATION HEALTH

$$ESS_k = (\sum_i w_i)^2 / \sum_i w_i^2 \quad ESS = N \text{ under uniform weights, } ESS = 1 \text{ when on}$$

THE $C_K = 0$ INVARIANT

Set every $C_k = 0$. Then $V \equiv 0, W = 1, R_k = 1, Z = 1$, and every final weight is 1, so the algorithm reduces to direct sampling exactly rather than approximately. A unit test pins that identity, and it is the strongest end-to-end check available on the scheme. The core is also validated against an analytic Ornstein–Uhlenbeck problem whose answer is known, with no GPU involved.

WHERE THE GUARANTEE WEAKENS

Eq. (5) is unbiased for a *deterministic* v . Standardizing the score across the live population, Eq. (2), is what lets one c_k schedule work across events with different units, and it makes v depend on the other walkers. The scheme is then self-interacting and the guarantee drops from **unbiased** to **consistent**. No closed-form confidence interval is available for the RES curve for the same reason. The band drawn in Fig. 2 treats the walkers as independent, which standardization makes them not, so it is a *lower bound* on the true uncertainty. The implementation accepts a fixed reference mean and standard deviation so that the strictly unbiased regime can still be tested.

4 · RESULT **The resampled population contains the event and direct sampling does not**

POPULATION		N	MEAN	SD	MAX	REACH +7.72 K
AI+RES	RESAMPLED	64	+8.24	1.49	+10.29	42
GenCast, Gate 3 walkers	DIRECT	16	+2.54	3.02	+7.49	0
GenCast, xres 0.25°	DIRECT	24	+2.46	2.66	+7.08	0
FCN3	DIRECT	24	+2.29	3.10	+7.87	1

Table 2. Realized A_L over the PNW box, in kelvin of 7-day-mean anomaly. The three direct rows are ensembles that already existed on disk, namely 16 free-running GenCast walkers from Gate 3, the 24-member 0.25° GenCast ensemble from the cross-resolution experiment, and the 24-member FCN3 ensemble from the head-to-head comparison. All four populations start from the same ERA5 analysis at the same init.

The headline claim needs no estimator. It is a statement about where walkers landed. 42 of 64 reach or exceed the ERA5 value and the population extends to +10.29 K, while the 40 direct GenCast members stop at +7.49 K. Nothing in that statement depends on a weight, a normalization, or a probability model.

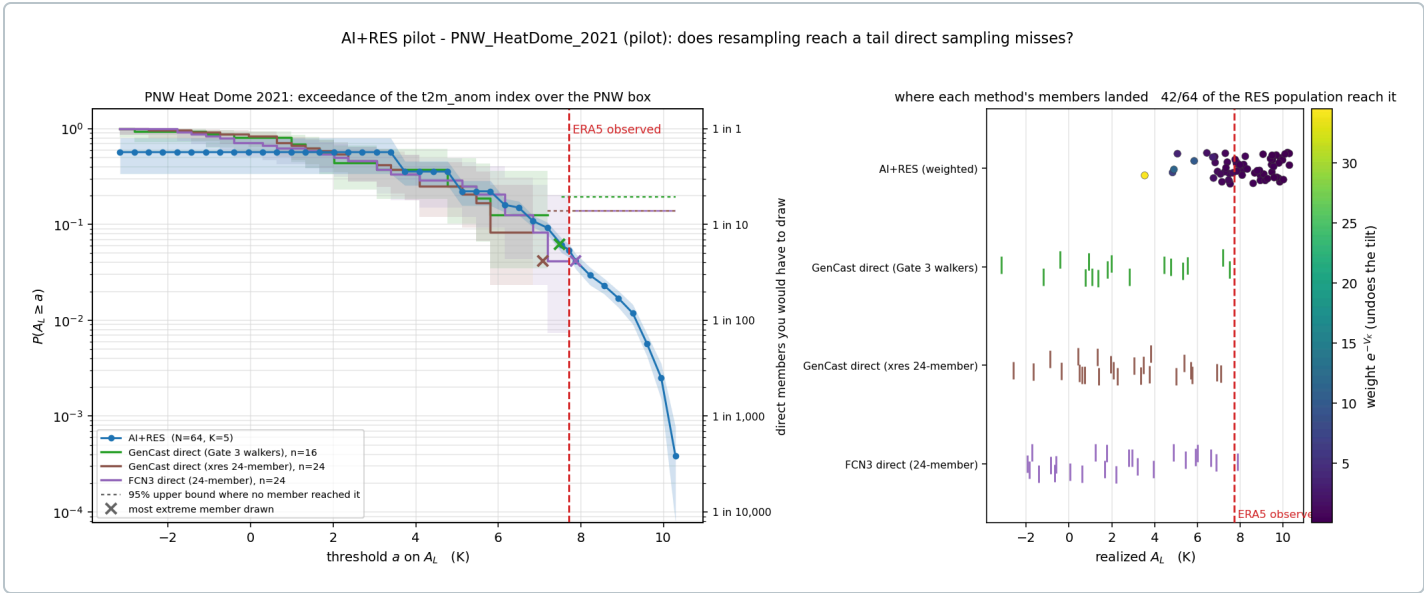


Figure 2. The tail as a probability. Left, the exceedance curve $P(A_L \geq a)$. AI+RES continues to 3.9×10^{-4} , one member in 2,600, where the direct ensembles run out of members and can report only a 95% upper bound. The right-hand axis converts the same curve into the number of members that would have to be drawn. Right, the same populations as raw values, colored by the weight that undoes the sampling tilt.

0.054 estimated $P(A_L \geq +7.72 \text{ K})$, about 1 in 19	3.9e-4 the far end of the curve, at +10.29 K, 1 in 2,600	0.14–0.19 the 95% upper bound available from all 40 direct draws	
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5 · IN SPACE

Whether the field is the event,
or only a matching box mean

A_L is an area mean, and an area mean can be reached by the wrong field. A ridge displaced a few hundred kilometres, or a broad shallow warm anomaly with no dome in it, scores the same as the event. Neither Fig. 1 nor Fig. 2 separates those cases. The maps do. Every panel in Fig. 3 to Fig. 7 is the same reduction as A_L on the same 105×237 grid, so walkers, direct members and observed truth are subtractable and share one colour scale.

AI+RES pilot - PNW_HeatDome_2021 (pilot): the observable as a FIELD, one shared scale.

The number under each panel is that panel's own box mean - the same A_L every score in this experiment is taken on. Black rectangle = the scored box.

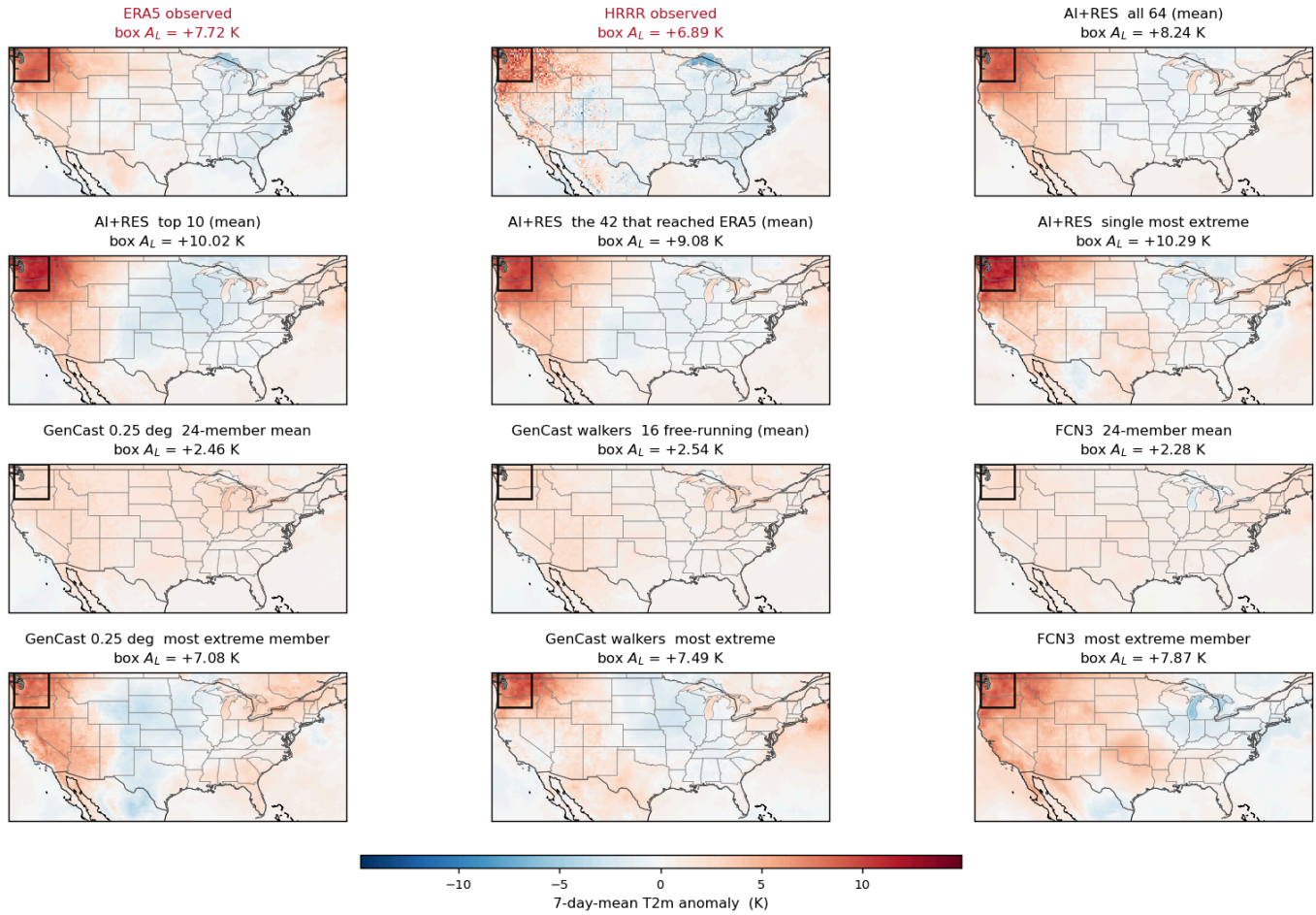


Figure 3. The observable as a field, one shared scale. Row 1 is truth plus the AI+RES population mean, row 2 the AI+RES tail, row 3 each baseline's *ensemble mean*, row 4 each baseline's *most extreme member*, chosen on its own box mean rather than on its hottest grid cell. Reading down a column therefore compares a baseline's mean with its own extreme. The number under each panel is that panel's box mean, the same A_L every score in this experiment is taken on. The black rectangle is the scored box.

At 21 days' lead the baseline *ensemble means* in Fig. 3 have no dome in them, between +2.3 and +2.5 K against an observed +7.72 K. Their most extreme members do have one, in the right place, and land at +7.08, +7.49 and +7.87 K. The pattern is available to a 24-member ensemble, and the amplitude is reached by 1 of the 64 direct members in Table 2. Every AI+RES panel is past the observed value.

AI+RES pilot - PNW_HeatDome_2021 (pilot): each field MINUS ERA5, one shared scale.

The first panel is HRRR minus ERA5 - two observational products disagreeing about the same week. A method whose error is no larger than that has not been shown to have the event in the wrong place.

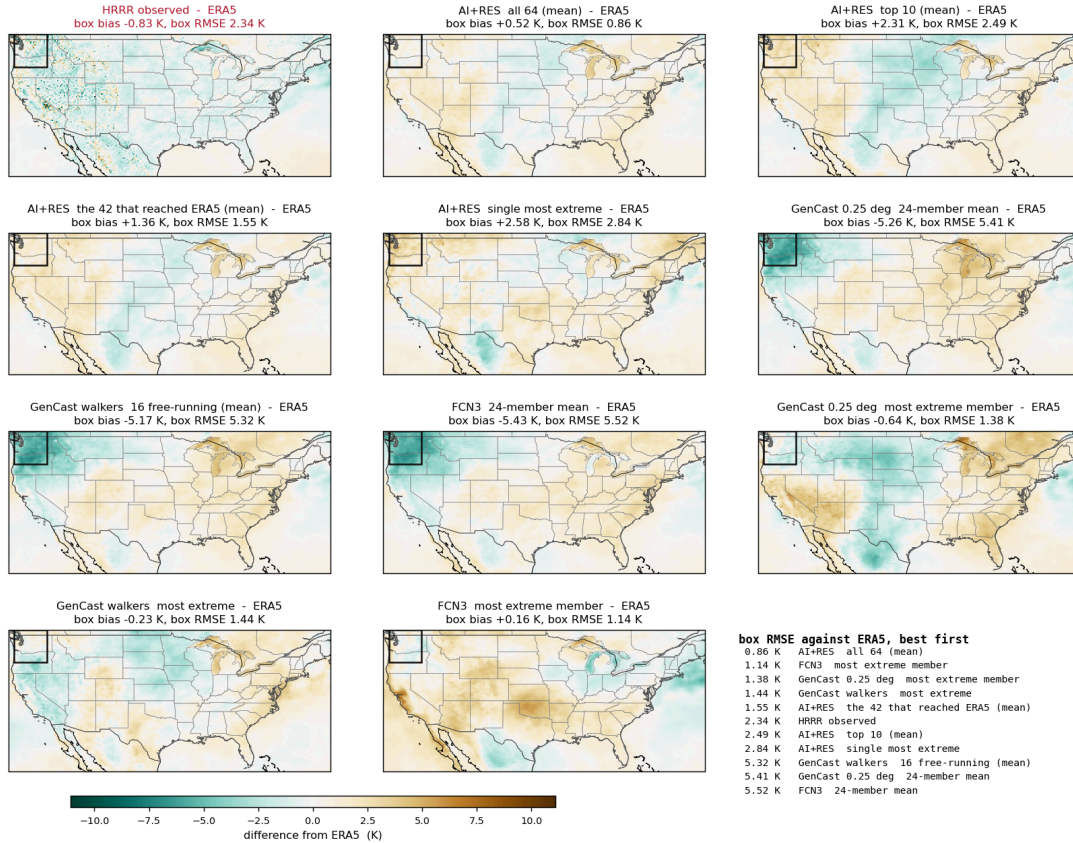


Figure 4. Each field minus ERA5. The first panel is **HRRR minus ERA5**, two observational products disagreeing about the same week by 2.34 K RMSE over the box. It is included as a reference scale, since a method whose error is no larger than that has not been shown to have the event in the wrong place. Teal is too cool, brown too warm.

FIELD	BOX BIAS	BOX RMSE
AI+RES, all 64 (mean)	+0.52 K	0.86 K
FCN3, most extreme member	+0.16 K	1.14 K
GenCast 0.25°, most extreme member	−0.64 K	1.38 K
GenCast walkers, most extreme	−0.23 K	1.44 K
AI+RES, the 42 that reached ERA5 (mean)	+1.36 K	1.55 K
<i>HRRR observed</i>	−0.83 K	2.34 K
AI+RES, top 10 (mean)	+2.31 K	2.49 K
AI+RES, single most extreme	+2.58 K	2.84 K
GenCast walkers, 16 free-running (mean)	−5.17 K	5.32 K
GenCast 0.25°, 24-member mean	−5.26 K	5.41 K
FCN3, 24-member mean	−5.43 K	5.52 K

Table 3. Box RMSE against ERA5 over the scored PNW box, best first. Bias is the box-mean difference. RMSE also counts pattern error, so a method can have near-zero bias and still be wrong in shape.

THE TOP ROW OF TABLE 3 IS NOT A SKILL CLAIM

The AI+RES population mean is the closest field to what happened, at **0.86 K**, inside the 2.34 K by which ERA5 and HRRR disagree. That population was aimed at this event, so its unweighted mean is not an estimate of the forecast mean and its closeness is not evidence of better forecast skill. What it does show is what a rare-event sampler is meant to deliver, that the observed event sits **inside** the resampled distribution rather than at its edge. A baseline ensemble mean, which is a forecast, is 5.2 to 5.5 K away.

The tail overshoots the event. The top-10 composite and the single most extreme walker carry +2.31 and +2.58 K of box bias in Table 3. The resampling was told to maximise A_L and it went past the observed value rather than stopping at it. That is the specified behaviour of the algorithm, and it means the far tail of this population is *more* extreme than 2021 rather than a reconstruction of it.

AI+RES pilot - PNW_HeatDome_2021 (pilot): the most extreme surviving lineage in space.
Contours are 500 hPa geopotential height (m) from the same walker - the ridge that carries the dome. The last panel is the 7-day mean the score is taken on.

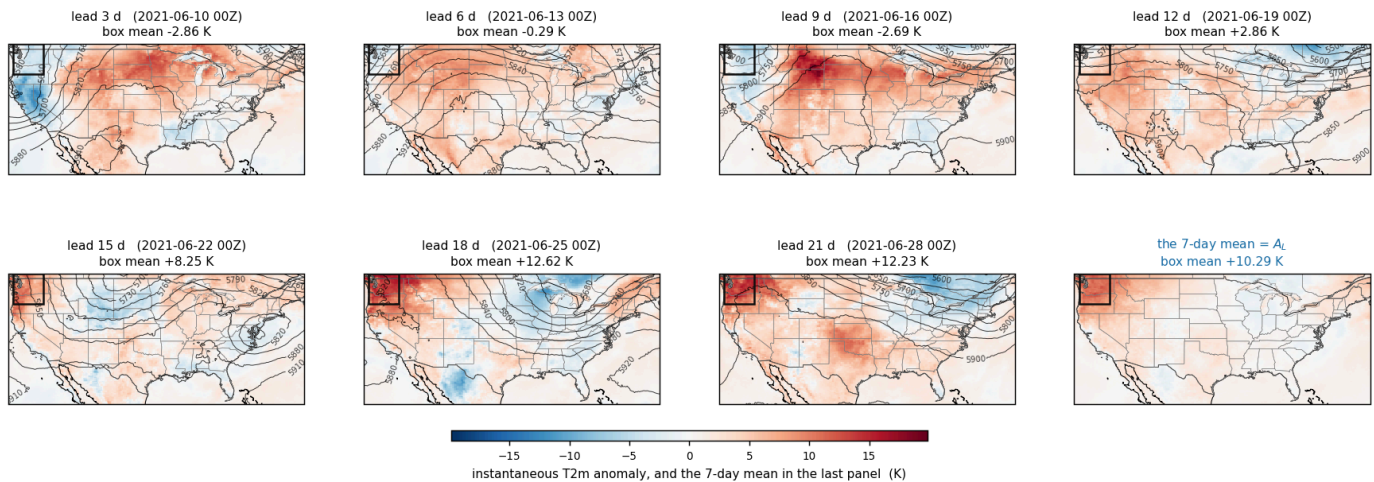


Figure 5. The most extreme lineage, in space. Slot 60, final $A_L +10.29$ K. Contours are 500 hPa geopotential height from the same walker, the ridge that carries the dome. Box mean by lead: -2.86, -0.29, -2.69, +2.86, +8.25, +12.62, +12.23 K. The last panel is the 7-day mean the score is taken on.

A SINGLE-WALKER CASE FOR AN AI SCORER

At the 9 d barrier this lineage occupied slot 63 and held the **4th-coolest current anomaly of the 64 walkers alive**, -2.69 K against a population mean of +1.46 K. That is $z = -1.33$ on a persistence score, which is under one expected child, so a persistence score would have killed it there. FCN3 ranked it 10th highest, $\theta = +7.18$ K, and it drew 11 children, the maximum multiplicity of that barrier. It became the most extreme walker of the run. This is one walker and therefore an illustration rather than evidence. The evidence is the aggregate in Table 4, where persistence scores $\rho_s = 0.06$ at that lead and FCN3 scores 0.80. The ridge at 9 d was already building over the Pacific while the surface under it had not yet warmed, so the score was reading the flow rather than the surface temperature.

AI+RES pilot - PNW_HeatDome_2021 (pilot): spread, and where the probability lives.
 Column 3 is the same estimator as the exceedance curve, per grid point: AI+RES uses the weights, the direct ensembles are raw member fractions and cannot go below one member. Column 2 is why AI+RES's 64 are worth fewer than 64: a tail-selected population of clones is TIGHTER than a free ensemble.

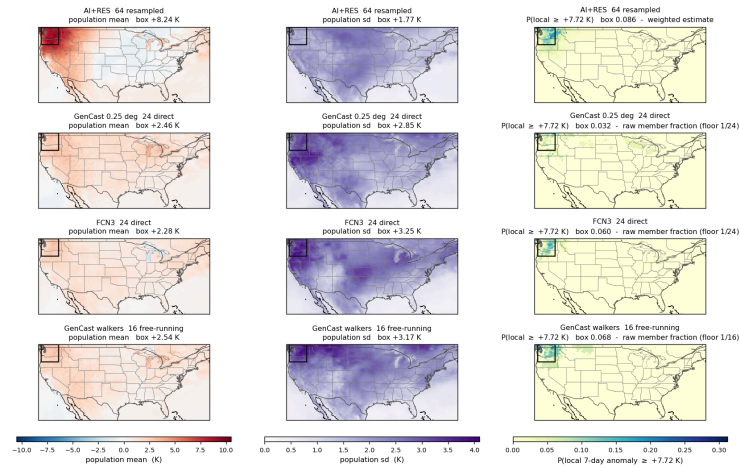


Figure 6. Spread, and where the probability sits. Column 3 is the exceedance estimator applied per grid point. AI+RES uses its weights, as the scalar curve of Fig. 2 does, and the direct ensembles are raw member fractions and cannot report anything below one member. The threshold in column 3 is *local*. A single grid point exceeding +7.72 K is easier than the whole box mean doing so, which is why the baselines are non-zero in column 3 while they are 0/24 on the box mean.

Column 2 of Fig. 6 is the cost of resampling. The AI+RES population's own spread over the box is 1.77 K against 2.85, 3.17 and 3.25 K for the three free ensembles. A tail-selected population of clones is *tighter* than a free one, which is the same finding Fig. 9 reports as a 60% separation deficit between siblings and unrelated pairs. It is why 64 resampled walkers are worth fewer than 64 independent draws.

AI+RES pilot - PNW HeatDome_2021 (pilot): what the resampling selected. Composites at the event peak and 3 days before. Z500 is the raw field: the only climatology on this machine is T2m, so the 'difference' column is the anomaly relative to the walker ensemble itself, not to a climate.

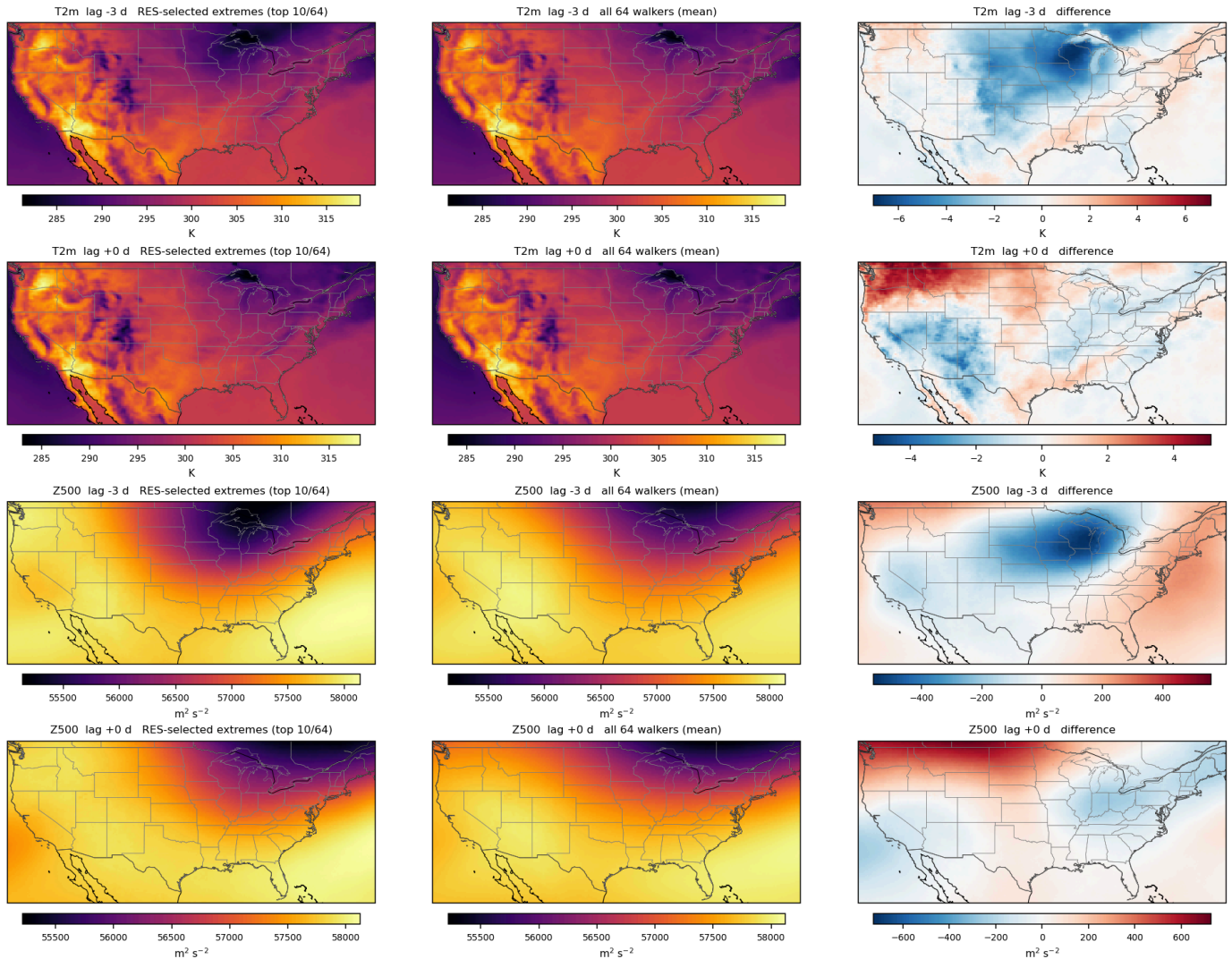


Figure 7. What the resampling selected. Left column, the composite of the **10 most extreme** walkers. Middle, the mean of all 64. Right, the difference. Rows are 2 m temperature and 500 hPa geopotential, each at the peak and 3 days before it. In the difference column, at the peak the selected walkers are 4 to 5 K warmer than the population over Oregon, Washington and British Columbia and *cooler* over the Great Basin, with a matching positive height anomaly on the same location. That is a ridge in the right place rather than a uniform warm bias, which is the failure mode a box-mean index cannot rule out on its own. As the figure itself notes, the only climatology on this machine is T2m, so the Z500 difference is relative to the walker ensemble and not to a climate.

6 • CAVEAT Two checks on the probability, with different verdicts

CHECK 1 • THE CURVES AGREE WHERE BOTH HAVE MEMBERS

Over the range +3.7 to +6.8 K, where the RES curve and the direct curves both have members, the ratio between them has a geometric mean of **1.27** and a range of **0.97 to 2.13**. That is inside the sampling error of 16 to 24 independent draws, whose Wilson intervals span factors of two to three over the same range. The RES curve is not systematically displaced.

CHECK 2 • THE TOTAL-MASS CHECK READS 0.568, NOT 1

An unbiased estimator assigns total probability 1 to everything. This run assigns **0.568**. That is ordinary at this ensemble size rather than a defect. Replaying the schedule 2,000 times at $N = 64$ puts 0.568 at the **33rd percentile**, and the statistic is right-skewed, with mean 0.99, *median 0.73*, standard deviation 1.17, and 36% of runs below 0.6. The value near 1 quoted in the estimator's own documentation is the mean, not what a single run at this N reads.

BOTH CHECKS HAVE THE SAME CAUSE, AND IT IS VISIBLE IN THE WEIGHTS

58 of the 64 final weights are below 2 and 52 are below 1. Four walkers, weighted 6.9, 10.0, 11.2 and 34.5, carry two thirds of the total mass. Rare-event resampling abandons the bulk by design. Only 11 of 64 founders still have descendants and no walker finished below +3.54 K, so the probability mass that belongs at low A_L is represented by a few heavily weighted survivors and $P(\text{everything})$ is noisy by construction. Below +3.54 K the RES curve in Fig. 2 is flat at 0.568 because it has no members there. That segment is the total-mass check plotted against a threshold axis, and it is not a bulk estimate.

Quote the ratios, not the absolute probabilities. Absolute probabilities from this run carry roughly a factor of two at $N = 64$. The ratio comparisons against direct sampling, and the reach of the population, do not.

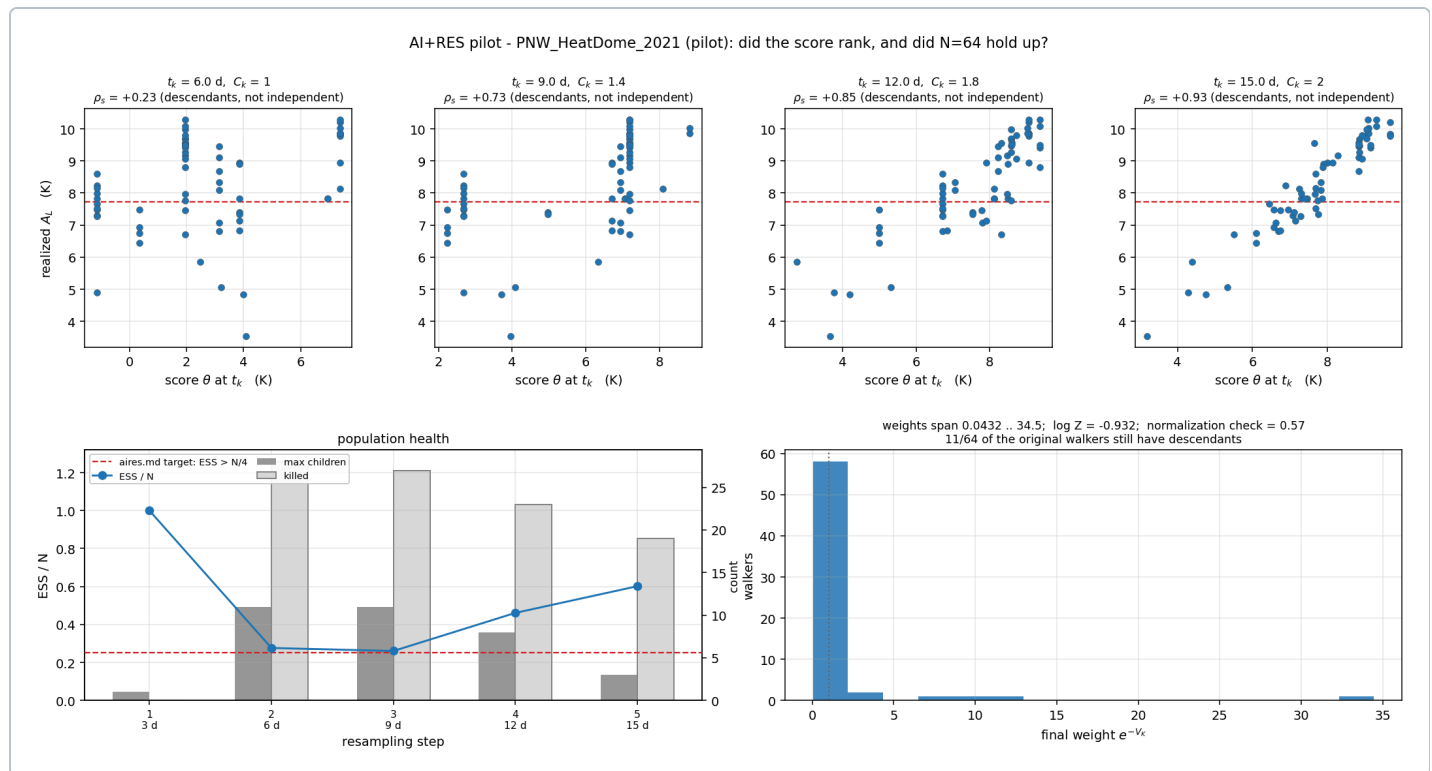


Figure 8. Score against outcome, barrier by barrier. Each panel is 64 walkers, the score θ received at that lead against the A_L realized 6 to 15 days later. Below, the effective sample size against the $N/4$ target, and the final weight distribution, 58 weights below 2 and four carrying most of the mass.

LEAD	PILOT P_S	GATE 3 P_S	PERSISTENCE P_S
3 d	n/a	-0.04	-0.38
6 d	+0.23	+0.61	+0.39
9 d	+0.73	+0.80	+0.06
12 d	+0.85	+0.96	+0.13
15 d	+0.93	+0.99	+0.56

Table 4. Rank correlation between the score at t_k and the walker's own realized A_L . The Gate 3 column is the same statistic on 16 *free-running* walkers, measured before the pilot was approved. The persistence column is the Standard-RES baseline of Lancelin et al., which scores a walker by how hot it already is. The pilot was not scored at 3 d, since $C_1 = 0$.

The pilot column of Table 4 reads lower than the Gate 3 column, and the cause is range restriction rather than a worse scorer. Gate 3's 16 walkers spanned -3.2 to +7.5 K. The resampled population spans +3.5 to +10.3 K and is concentrated above +7 K. Correlation is attenuated by that compression. These walkers are also

descendants of one another rather than independent draws, so the number is not comparable to the gate's.

WHY AN AI SCORER RATHER THAN THE CHEAP ONE

The Standard-RES of Lancelin et al. scores a walker by the observable's current value, which costs nothing. At the two leads the gate was taken on, that baseline has no skill, 0.06 at 9 d and 0.13 at 12 d, both inside the noise, while FCN3 reaches 0.80 and 0.96 in Table 4. Persistence becomes useful only at 15 d, by which point the verification window has begun. FCN3 is therefore not reading off the current state, and that is what the GPU hours buy.

8 • GENEALOGY Eleven founders, and clones that separate on their own

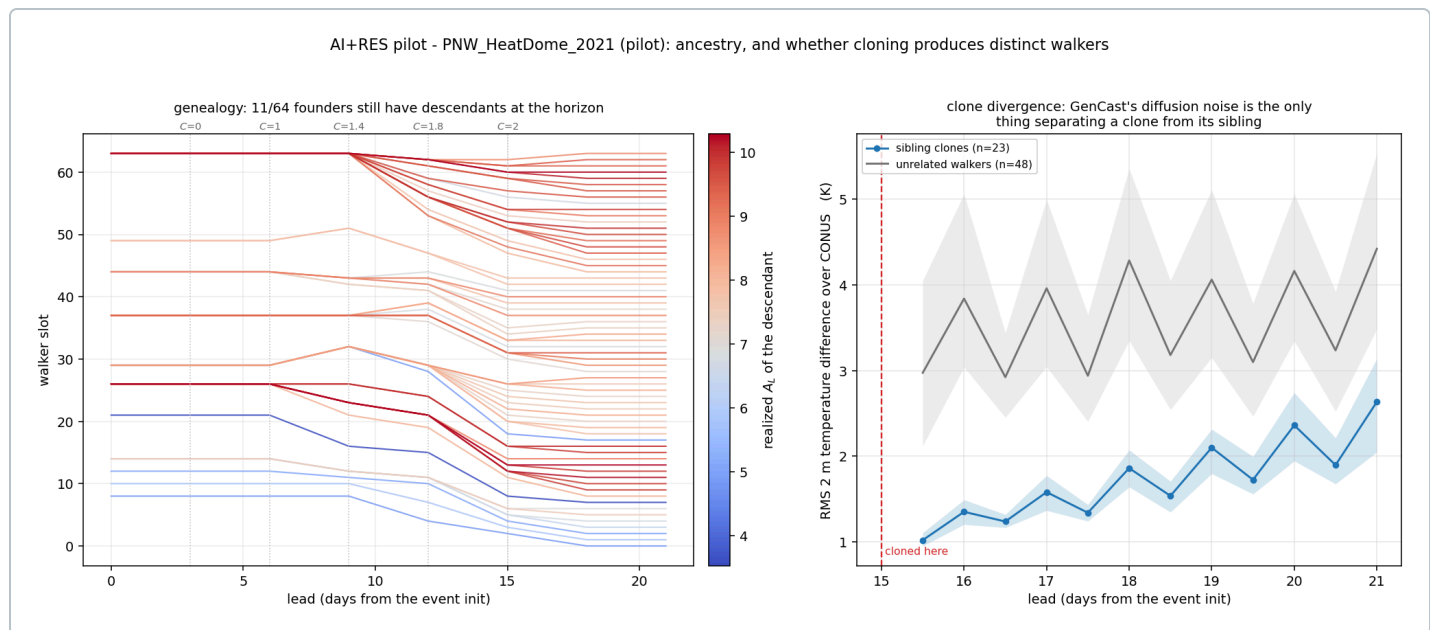


Figure 9. Left, the tree of Fig. 1 indexed by slot rather than tinted by temperature, which shows which slot continued which and the collapse onto 11 surviving founders. Right, the design bet.

Lancelin et al. perturb log surface pressure with spherical harmonics because PlaSim is deterministic and its clones would otherwise stay bit-identical. GenCast's diffusion noise was expected to make that perturbation unnecessary.

WHAT THE SIBLING SEPARATION SHOWS

- Sibling clones born at the 15 d barrier differ by **1.0 K RMS** in 2 m temperature over CONUS after a *single* 12 h step, growing to about **2.5 K** by the 21 d horizon. No hand-tuned perturbation was needed.
- Unrelated walkers sit at **3.0 to 4.5 K** over the same span. At the horizon siblings are separated by about **60%** of the unrelated-pair distance, so **clones remain correlated at verification** and the effective number of independent tail samples is below 64.
- This is the first quantitative measurement of that correlation, and it is the argument for adding the perturbation of Lancelin et al. if a later run resamples harder or later than this one did.

9 · CONFIG Every knob, and what it was set to

The driver freezes its configuration to `run.json` before the first walker moves and refuses to resume a run whose configuration has changed, so two runs that disagree on any of this are two runs. Table 5 is read from that frozen record, or from the module constants it does not capture.

KNOB	VALUE	WHAT IT DECIDES
THE WALKER, GENCAST		
params_file	GenCast 0p25deg <2019.npz	Trained on pre-2019 data only, so a 2021 event is out of sample. Occupies about 64 GB of an 80 GB H100, which is why every walker step is serial.
resolution	0p25	0.25°, global. Walker states must be global, since a CONUS crop cannot restart a walker or initialise the scorer.
step_h	12	GenCast's native step. 42 steps from the init to the peak.
seg_steps	6	6 × 12 h = one 3-day segment, the unit of work handed to a worker. One XLA compile per worker per leg.
n_walkers	64	Population size N. Sets the resolution of the tail and the factor-of-two uncertainty on the probability.
horizon_days	21.0	The peak. Must equal the init-to-peak lead, otherwise the walkers never enter the verification window and A_L is undefined.
keep_states §	2	Segments of walker state kept resident. Each state is 477 MB, which is what held peak disk at 37 GB.
THE SCORE, FCN3		
backend	fcn3	The alternative, persistence, is the Standard-RES baseline of Lancelin et al. and was run as a separate control.
members	6	M, the score ensemble. θ is the mean of these 6 members' A_L .
FCN3 step §	6 h	The scorer's native step, half the walker's. A barrier at 6 d forecasts 60 steps to the peak, and one at 15 d forecasts 24.
IC perturbation §	Zero()	All 6 members start from the same 72-channel field. Their spread is FCN3's own stochasticity rather than a perturbed initial condition.

Knob	Value	What it decides
skip_null_scores	true	A barrier with $C_k = 0$ resamples nothing, so its score is never computed. Barrier 1 therefore cost zero GPU hours.
THE RESAMPLING, DIFFUSION MONTE CARLO		
C	0, 1.0, 1.4, 1.8, 2.0	Selection pressure per barrier. Clone count $\propto \exp(C_k \cdot \text{standardized score})$. The score is standardized across the live population at each barrier, which makes the scheme self-interacting and rules out a closed-form error bar, as Section 3 states.
leads	3, 6, 9, 12, 15 d	Where the barriers sit. The last is 6 days before the peak, so the event window is a free run-out.
tail_sign	+1.0	HIGH tail, so the largest A_L is cloned. A cold event such as Uri sets -1 and the same code selects the most negative.
ESS target $\S$	$> N/4 = 16$	The health floor, not an input. Breaching it means the population has collapsed onto too few ancestors to estimate anything.
THE OBSERVABLE		
metric	t2m_anom	Departure from the 1990–2019 climatology, sampled at each frame’s own hour and day-of-year.
box $\S$	44–50 N, 124–118 W	Oregon and Washington. A CONUS mean of this event is mostly grid points that were not having a heat wave. The 50 N edge is the CONUS crop rather than a choice.
window $\S$	13 frames, 15→21 d	The 7-day mean ending at the peak. check_window hard-fails on any other frame count rather than averaging over two frames and calling it A_L .
climatology $\S$	105 × 237 @ 0.25°	Verified by grid rather than by filename, after a 2° file of the same name on another machine produced silently wrong anomalies.
SEEDS		

Knob	Value	What it decides
base_seed	20260101	Walker noise. The stream is <code>fold_in(fold_in(key, slot), segment)</code> , keyed on the <i>slot</i> , so a clone draws a different stream from its sibling. That is what makes clones separate.
dmc_seed	20260819	The resampling draw. Fixing it makes the genealogy replayable from the cached scores in seconds.
score seed §	CRC32(event, lead)	Not a function of the walker. Every walker at one barrier is scored under the same FCN3 noise, so a difference in θ is a difference in state. Common random numbers.
Runtime		
shards §	8	One process per H100 on one node, for both the walk and the score pools.
attempts §	3	Per-leg retries. A leg that loses a node to a foreign process retries rather than killing the run, and cached segments skip.

Table 5. Configuration frozen at `runs/aires/PNW_HeatDome_2021/res/pilot/run.json`. Values marked § are not in the frozen block. They come from `aires/aconfig.py`, `aires/aindex.py` or the Slurm script, and are recorded here because they change the answer.

TWO SETTINGS WERE CHOSEN AGAINST A MEASUREMENT

$M = 6$ and the c_k ramp both come from work done before the run. c_k was replayed on Gate 3's measured skill curve and predicted the effective sample sizes of Section 2 to within a few hundredths. M is a floor set by the noise on an ensemble mean. At $M = 6$ two ensemble means of this event's A_L differ by 0.36 K from sampling noise alone, which is adequate for *ranking* walkers and was fatal to a 0.25 K equivalence threshold, the reason Gate 2 was re-scored at $M = 24$.

LEG	LEAD	WALK	SCORE	0 MEAN
1	0 → 3 d	40.3 min	skipped	-3.18
2	3 → 6 d	34.4 min	71.8 min	+1.98
3	6 → 9 d	41.4 min	55.7 min	+4.07
4	9 → 12 d	42.0 min	43.2 min	+5.57
5	12 → 15 d	39.0 min	30.6 min	+6.62
6	15 → 21 d	64.9 min	n/a	n/a

Table 6. One 8×H100 node, 465 min of coordinator wall, against the 7.6 h the corrected rate estimates predicted. Leg 1 rolled only 48 of 64 segments because 16 were already on disk from Gate 3, and its score was skipped because $c_1 = 0$, so its entry in the last column is the walkers’ own box mean printed by the driver rather than an FCN3 score. Leg 6 carries the population through the event window and rolls 128 segments with no barrier at the end.

Peak disk was **37 GB** against 92 GB projected. With only two segments of walker states ever resident, the projection had been charging for three. All 448 diagnostics cubes were retained, which is why Fig. 1 could be drawn, including its killed branches, months after the run.

THE STACK, IN ONE PARAGRAPH

GenCast is the **walker**, the object propagated, cloned and killed. FCN3 is the **score function**. At each barrier it is handed the walker’s state and asked, with 6 members run to the peak, where that walker is heading. The roles are inverted relative to the obvious reading of the two models, in that the diffusion model does the physics and the deterministic-core model does the judging. They cannot share a process, JAX against PyTorch, so walker and scorer hand off through disk, and the join is a 72-channel adapter in which 69 channels are a rename plus a latitude flip and 3 are derived from a fitted calibration.

11 · PROVENANCE

The three gates this run had to pass first

GATE	WHAT IT ASKS	MEASURED	THRESHOLD	
1	Do the adapter’s 3 derived channels match a native FCN3 IC?	0.63 m/s · 4.6%	1.0 m/s · 5%	PASS
2a	Does an adapter IC give the same ensemble-mean forecast?	0.083 K	< 0.25 K	PASS
2b	Out to what lead are the two ensembles rank-equivalent?	8.5 d	≥ 7 d	PASS
2c	Is the adapter’s perturbation smaller than FCN3’s own noise?	0.990	≤ 1.00	PASS
3	Does the score rank free-running walkers at 9 d and 12 d?	0.80 · 0.96	≥ 0.50	PASS

Table 7. Gates, each specified with a threshold before it was measured. Gate 2’s second criterion was amended from a 21-day endpoint to a horizon test, and that amendment is the substantive result of that gate. At 21 d lead any initial-condition difference has been amplified to the internal-noise level, so the original criterion could not have been passed by a perfect adapter either.

A separate ten-case follow-up asked whether the adapter costs forecast skill against observed truth, using six extreme events and four quiet-day null controls. The pooled effect is **+0.23%** with a 95% interval of [−5.20, +5.65] and no regime dependence, so whatever the adapter costs at week 3 lead is under about 5% of the native run’s RMSE. That bound is the number that matters for AI+RES. The walker score is an ensemble mean, and a bias this far under the ensemble’s own sampling noise cannot reorder walkers.

12 · OPEN

What this pilot does not settle

- **One event, one atmosphere.** Everything here is the 2021 PNW dome from one init. The reach claim is robust, and the probability is one draw of one schedule at $N = 64$.
- **Clones are not independent.** A separation at 60% of the unrelated-pair distance at verification means the 64 tail samples are worth fewer than 64. Quantified, not yet fixed.
- **The box clips the event.** The observed dome maximum reached about 52 N into British Columbia, outside the 50 N CONUS crop the cubes carry, so the scored box sees the southern two thirds of what happened.
- **The far tail overshoots the event.** The top-10 composite carries +2.31 K of box bias and the single most extreme walker +2.58 K. That is correct for an algorithm

told to maximise A_L , and it means the extreme end of this population is more extreme than 2021 rather than a reconstruction of it.

- **Absolute probabilities need more walkers.** The factor-of-two uncertainty on P is an N problem, and the total-mass diagnostic is the statistic to watch as N grows.

AI+RES phase 4 pilot · PNW_HeatDome_2021, init 2021-06-07, peak 2021-06-28 · $N = 64$ walkers, $M = 6$ score members, $c_k = (0, 1.0, 1.4, 1.8, 2.0)$ · Slurm job 1172 on one 8×H100 node, 465 min · $\log Z = -0.9317$, total-mass check 0.5684, 11/64 founders surviving.

Figures regenerated from `runs/aires/PNW_HeatDome_2021/res/pilot/` by `aires/aplots.py`. Method after Lancelin et al., [arXiv:2510.27066](https://arxiv.org/abs/2510.27066), with the roles inverted, so GenCast walks and FCN3 scores.